

1 What is Cucumber?

Answer:

Cucumber is a **BDD (Behavior Driven Development) testing framework** that allows writing test cases in a **human-readable format** (Gherkin language). It bridges the gap between **business stakeholders and technical teams**.

2 What is BDD?

Answer:

BDD stands for **Behavior Driven Development**.

- It focuses on **what the application should do** from the user's perspective.
- Uses **Gherkin language** (**Given**, **When**, **Then**) for writing scenarios.
- Example:

Given(precondition) the user is on the login page

When(action) the user enters valid credentials

Then(outcome) the user should see the dashboard

3 What is Gherkin?

Answer:

Gherkin is the **syntax/language** used in Cucumber to write test cases.

- Keywords: **Feature**, **Scenario**, **Given**, **When**, **Then**, **And**, **But**.
- Example:

Feature: User Login

Scenario: Successful login

Given the user is on the login page

When the user enters valid credentials

Then the user should see the dashboard

4 What are Feature Files in Cucumber?

Answer:

- A **feature file** contains **all scenarios** for a feature in Gherkin syntax.
- Extension: `.feature`
- Example:

Feature: User Login

Scenario: Successful login

Given the user is on the login page

When the user enters valid credentials

Then the user should see the dashboard

5 What are Step Definitions in Cucumber?

Answer:

- Step definitions contain **the actual Selenium or Java code** for each step in a scenario.
- Each Gherkin step is mapped to a method with **annotations**: `@Given`, `@When`, `@Then`.

Example:

```
@Given("the user is on the login page")

public void user_on_login_page() {

    driver.get("https://example.com/login");
}
```

6 What are Hooks in Cucumber?

Answer:

- Hooks are **special methods** that run **before or after each scenario**.
- Annotations: `@Before`, `@After`
- Example:

`@Before`

```
public void setup() {  
    driver = new ChromeDriver();  
}
```

`@After`

```
public void teardown() {  
    driver.quit();  
}
```

7 What is a Scenario in Cucumber?

Answer:

- A scenario describes **one test case** in a feature file.
- Steps: `Given`, `When`, `Then`

Example:

Scenario: Login with valid credentials

Given the user is on the login page

When the user enters valid credentials

Then the user should see the dashboard

8 What is a Scenario Outline?

Answer:

- Scenario Outline is used when the **same scenario should run with multiple sets of data**.
- Uses **Examples** table.

Example:

Scenario Outline: Login with multiple users

Given the user is on the login page

When the user enters username "<user>" and password "<pass>"

Then the user should see the dashboard

Examples:

user	pass	
test1@gmail.com	12345	
test2@gmail.com	54321	

9 What is the difference between Scenario and Scenario Outline?

Scenario	Scenario Outline
Runs once with specific data	Runs multiple times with multiple sets of data
No Examples table	Requires Examples table

10 What are Tags in Cucumber?

Answer:

- Tags are used to **group scenarios** and **run selective tests**.
- Example:

@Smoke

Scenario: Login test

Given the user is on the login page

- Run specific tag:

```
mvn test -Dcucumber.options="--tags @Smoke"
```

11 What is Glue Code in Cucumber?

Answer:

- Glue code is the **package that contains step definitions**.
- Mentioned in @CucumberOptions in the test runner.

Example:

```
@CucumberOptions(features="src/test/resources/features",  
glue="stepDefinitions")
```

12 What is the difference between Feature File and Step Definition File?

A Feature file is written in Gherkin language and is human-readable, containing test scenarios described using Given, When, and Then statements to define what needs to be tested.

In contrast, a Step Definition file is written in Java and contains the actual automation code that executes those steps, linking the Gherkin statements from the feature file to Selenium or other automation actions.

13 What is the role of Test Runner in Cucumber?

Answer:

- Test Runner is a **Java class** that runs Cucumber tests.
- Uses `@RunWith(Cucumber.class)` and `@CucumberOptions`.

Example:

```
@RunWith(Cucumber.class)

@CucumberOptions(

    features = "src/test/resources/features",

    glue = "stepDefinitions",

    plugin = {"pretty", "html:target/cucumber-report.html"})

public class TestRunner {}
```

14 What are Data Tables in Cucumber?

Answer:

- Data Tables allow passing **multiple data values** to a step.
- Example:

Given the user fills the form with data

Name	Email
Alice	alice@mail.com

Step definition:

```
@Given("the user fills the form with data")

public void fill_form(DataTable table) {

    List<Map<String,String>> data = table.asMaps(String.class,
String.class);

    driver.findElement(By.id("name")).sendKeys(data.get(0).get("Name"));

}
```

15 Difference between Cucumber and Selenium

Selenium	Cucumber
Automation tool for web apps	BDD framework
Executes tests	Writes test scenarios
Works with code	Works with Gherkin language
Can be used without Cucumber	Needs Selenium/other frameworks to automate

16 Explain BDD vs TDD.

TDD (Test-Driven Development) means writing *test cases first* before writing the actual code. The focus is on the developer side.

Developers write a test → code until it passes → refactor.

BDD (Behavior-Driven Development) focuses on *behavior from a user or business point of view*.

Here, test cases are written in **plain English (Gherkin language)** like:

“Given user logs into CBS, When user renews FD, Then confirmation message is shown.”

BDD helps testers, developers, and business teams understand tests easily.